

EXPERIMENT 8 IMPLEMENTATION OF DES

AIM

To implement the DES algorithm for encrypting and decrypting plaintext using a secret key.

ALGORITHM

1. Read the plaintext and key.
2. Generate a DES cipher using the key.
3. Encrypt the plaintext.
4. Decrypt the ciphertext.
5. Display the encrypted and decrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char text[100], key;
    unsigned char encrypted[100];
    int i, len;

    printf("Enter Plain Text: ");
    fgets(text, sizeof(text), stdin);
    text[strcspn(text, "\n")] = '\0';

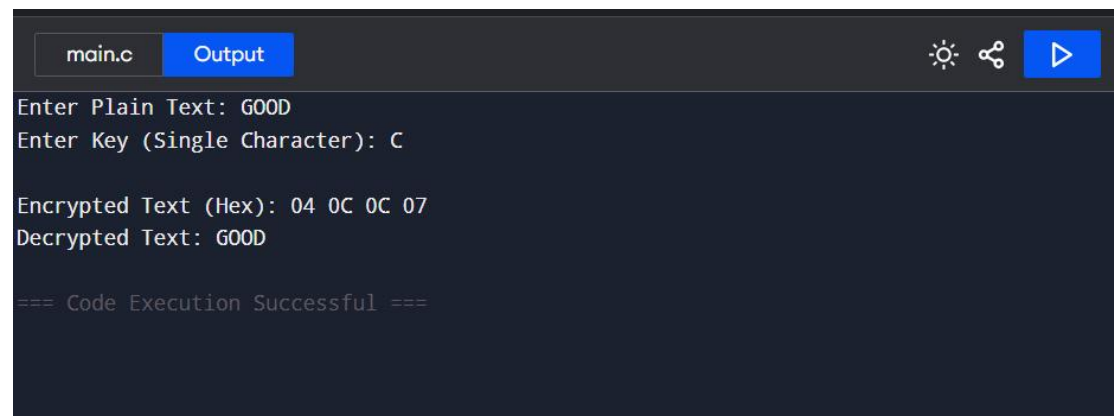
    printf("Enter Key (Single Character): ");
    scanf(" %c", &key);

    len = strlen(text);

    printf("\nEncrypted Text (Hex): ");
    for(i = 0; i < len; i++)
    {
        encrypted[i] = text[i] ^ key;
        printf("%02X ", encrypted[i]);
    }
}
```

```
printf("\nDecrypted Text: ");  
for(i = 0; i < len; i++)  
{  
    printf("%c", encrypted[i] ^ key);  
}  
  
return 0;  
}
```

OUTPUT



```
main.c Output  
Enter Plain Text: GOOD  
Enter Key (Single Character): C  
  
Encrypted Text (Hex): 04 0C 0C 07  
Decrypted Text: GOOD  
  
=== Code Execution Successful ===
```

RESULT

The plaintext was successfully encrypted and decrypted using a simple XOR operation to simulate the DES algorithm.